Field-Robust Paddy and Crop Disease Recognition with Lightweight Vision Transformers for Edge Deployment

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***Abstract*—Crop disease classifiers trained on clean images lose accuracy in the field, and the models that cope best with field images are often too heavy for a phone. This paper asks how much of a lightweight vision transformer's accuracy a MobileNetV3 student can recover through distillation when the student is trained on field-style distorted images. The teacher (MobileViT-S for PlantDoc, EfficientFormer-L1 for Paddy) sees the clean image, while the student sees a blurred, rotated, occluded and re-lit copy of it. Experiments use PlantDoc, whose images come from uncontrolled settings, and a 10-class subset of Paddy Doctor collected in Tamil Nadu paddy fields, with grouped splits that keep near-duplicate photographs together. On PlantDoc, distillation raises the student's macro-F1 from 56.0 ± 1.7 to 62.1 ± 1.0 over three seeds, closing about 40% of the gap to the teacher (71.1) with 6.6 times fewer operations. On the in-distribution Paddy split all models reach 93–95% accuracy, and neither the augmentation nor the distillation helps. When trained on one rice variety and tested on the others, every model drops to 29–35 macro-F1 and the distilled student gains nothing. Mixing feature statistics (MixStyle) and tuning the distillation weight do not change this: the PlantDoc differences stay within seed spread, and the student's variety-shift score stays at 30–32. Robustness to unseen varieties remains an open problem.**

***Index Terms*—plant disease recognition, lightweight vision transformer, knowledge distillation, domain shift, edge deployment**

I. INTRODUCTION

Plant disease models that work well on clean, lab-style photographs often fail on pictures taken in a field. The authors of PlantDoc report that India loses 35% of its annual crop yield to plant diseases and that lab-controlled datasets cannot be used to classify real-scenario images [1]. Later studies describe the same lab-to-field shift and try to reduce it with domain adaptation and cross-domain few-shot learning [2], [3]. A model meant for farmers also has to run on a phone, which favours small networks such as MobileNetV3 [4], [5] and lightweight vision transformers such as MobileViT [6] and EfficientFormer [7].

Lightweight transformers recover accuracy that small CNNs lose on hard field images, but they cost more to run. In the measurements reported in Section VI, MobileViT-S needs 6.6 times more operations than MobileNetV3-Large and runs about three to four times slower on a CPU. Knowledge distillation [8] is a natural way to move the transformer's accuracy into the cheaper CNN, and recent plant-disease work already distils into MobileNet-style students [9]–[13]. Much less is known about whether such a student stays accurate when the field conditions or the crop variety change.

Two questions guide the study. **RQ1:** how much of a lightweight ViT's accuracy on field images can a MobileNetV3 student recover through distillation under field-style distortion? **RQ2:** does the recovered accuracy survive a change of rice variety?

The contributions are as follows. First, a controlled comparison on two field datasets (PlantDoc and a 10-class Paddy Doctor subset) of distillation from a lightweight ViT teacher, in which the teacher sees the clean image and the student sees a distorted one, with ablations over three seeds on PlantDoc. Second, a variety-shift protocol for paddy (train on variety ADT45, test on the other nine) that exposes a large robustness gap: every model falls from above 90% to 34–42% accuracy. Third, near-duplicate-aware splits, per-class metrics and measured cost (parameters, operations and CPU latency), with code released for reproduction.

No new distillation loss is proposed. The aim is to adapt a standard one to field images and to measure carefully where it helps and where it does not.

II. RELATED WORK

Datasets. PlantDoc contains 2,598 images of 13 plant species in 27 classes, all collected outside the lab, and its authors found that fine-tuning on it cut classification error by up to 31% compared with lab-trained models [1]. Paddy Doctor provides 16,225 smartphone images of paddy leaves in 13 classes from Tamil Nadu fields [14]; a 10,407-image subset was used in a public Kaggle competition. Dhan-Shomadhan offers 1,106 rice-leaf images with field and white backgrounds [15].

Lightweight networks. MobileNetV3 is a small CNN found by architecture search [4]. MobileViT combines convolutions with transformer blocks for mobile devices [6], and EfficientFormer redesigns a pure transformer to reach MobileNet-like latency [7]. All three serve as baselines in this study.

Lightweight and hybrid ViTs for plant and rice disease. PMVT modifies MobileViT with an inverted residual block and a CBAM attention module, and is tested on wheat, coffee and rice data [16]. MobilePlantViT is a hybrid ViT for generalized plant disease classification on mobile devices [17]. For rice, recent work combines EfficientNet-B0 with a

ViT [18], a ViT with ResNet18 [19], several ViTs fused by decision [20], and a lightweight MaxViT-based network for crop diseases [21]. A mobile application study uses the Paddy Doctor data with YOLOv8 and a ViT [22]. These papers mostly measure in-distribution accuracy.

Edge deployment. Khan et al. build a MobileNetV3-based plant disease model exported to ONNX for edge devices and evaluate it on PlantVillage [5].

Distillation in agriculture. The standard soft-label formulation is due to Hinton et al. [8]. A multistage distillation method improves a lightweight detector and reaches 60.4% mAP@.5 on PlantDoc [9]. Ensemble self-distillation improves ShuffleNetV2 on tomato leaf diseases [10]. A hybrid distillation framework yields a student of about 1.07M parameters for rice seed variety and leaf disease classification [11]. Two very recent works are closest to this one: AgriKD distils a ViT teacher into a compact convolutional student for edge leaf disease classification [12], and a 2026 preprint distils into a MobileNetV3 student with category-guided knowledge [13]. Both concentrate on accuracy and efficiency; judging from their abstracts, neither evaluates robustness to a change of crop variety.

Domain shift. Models trained under one acquisition setting lose accuracy in another. Wu et al. address lab-to-field shift with unsupervised domain adaptation [2], and Yang et al. show that few-shot models degrade sharply on field datasets [3]. MixStyle [23] mixes channel-wise feature statistics between training images to improve generalisation to unseen domains, and it is tested here inside the distilled student.

Gap. Distillation into small students is well established, and so are lab-to-field studies, but few works test whether a distilled lightweight student survives a change of variety within a single crop. This paper offers a controlled measurement of exactly that, using an asymmetric setup in which the teacher sees the clean image and the student a distorted one.

III. DATASETS AND PREPROCESSING

Two field datasets are used: PlantDoc (cropped classification version), with 2,576 images in 27 classes [1], and the Paddy Doctor Kaggle subset, with 10,407 smartphone photographs (480×640) of paddy fields near Tirunelveli, Tamil Nadu, in 10 classes [14]. The full Paddy Doctor release (16,225 images, 13 classes) requires an IEEE DataPort subscription and could not be accessed, so the 10,407-image Kaggle competition subset of the same collection was used instead.

Paddy Doctor subset. The ten classes are bacterial leaf blight (479 images), bacterial leaf streak (380), bacterial panicle blight (337), blast (1,738), brown spot (965), dead heart (1,442), downy mildew (620), hispa (1,594), normal (1,764) and tungro (1,088). The largest class is about five times the smallest. Each image carries a rice variety and a crop age (45–82 days, mean 64). Variety ADT45 makes up 6,992 images (67%), and variety and disease are correlated: AtchayaPonni, for example, supplies the largest share (340 of 761) of the non-ADT45 hispa images, and 83% of normal leaves are ADT45. The grouped split holds 7,457 training, 1,506 validation and 1,444 test images.

PlantDoc. The official train and test folders of the cropped classification version are used. The training folder contains one class (two-spotted spider mites, 2 images) that has no test images, so it was dropped, leaving the standard 27 classes. Fifteen percent of the training folder, stratified by class, became the validation set, which gives 1,989 training, 351 validation and 236 test images. The official test set was left untouched.

Leakage control. Paddy photographs are often several shots of the same leaf. A perceptual hash (pHash) was computed for every image and images within Hamming distance 4 were linked, which produced 8,442 groups; 2,978 images (29%) sit in groups of two or more, the largest with 25 images. A stratified group split keeps each group inside one split, so near-duplicates never straddle train and test.

Preprocessing. For speed, all images were resized once so that the longest side is 384 px (PlantDoc) or 320 px (Paddy), and every model sees the same copies. During training the images are resized to 224×224 and normalised with ImageNet statistics.

Variety-shift split (Paddy only). Training and validation use only the ADT45 images from the grouped split. The test set is made of all 3,415 images of the other nine varieties. All ten disease classes appear in it, the smallest with 73 images.

IV. METHOD

The student is a MobileNetV3-Large pretrained on ImageNet. It is trained with a weighted cross-entropy loss and, optionally, a distillation term from a frozen lightweight-ViT teacher and a field-style augmentation pipeline. Only the student is used at test time.

Field-style augmentation. The baseline pipeline uses random-resized-crop (scale 0.5–1.0), horizontal flip and colour jitter (brightness, contrast and saturation 0.3). The field-style pipeline widens the crop (scale 0.35–1.0) and adds rotation up to 25°, stronger lighting changes (brightness and contrast 0.5, saturation 0.4), Gaussian blur (kernel 5, sigma 0.1–2.0, probability 0.3) and random erasing (probability 0.5, 2–20% of the image). These imitate shadows, motion blur, tilted leaves and occlusion by other leaves.

Asymmetric distillation. For each training image the teacher receives the clean view x (resized to 224×224) and the student receives a distorted view $\tilde{x}$. The student learns from the labels and from the teacher's softened prediction on the clean view:

$$L = (1 - \alpha)\, CE_w(z_s(\tilde{x}), y) + \alpha\, T^2\, KL(\sigma(z_t(x)/T) \,\|\, \sigma(z_s(\tilde{x})/T)) \quad (1)$$

Here z_s and z_t are the student and teacher logits, σ is the softmax, and CE_w is cross-entropy with label smoothing 0.1 and class weights proportional to the square root of (largest class count / class count), normalised to a mean of 1. The main results use $\alpha = 0.5$ and $T = 4$; a validation-based sweep

of α is reported in Section VI. The intuition is that the teacher's judgement of the clean image tells the student what the label should look like even when the student's own view is degraded.

Teachers. Each teacher is fine-tuned with the baseline recipe before distillation: MobileViT-S for PlantDoc and EfficientFormer-L1 for Paddy. In the variety-shift experiment the teacher is retrained on ADT45 images only, so the test varieties cannot leak into the student through the teacher.

Style mixing (optional). For the variety-shift question, MixStyle [23] was also tried in the student. With probability 0.5 per batch, the channel-wise mean and standard deviation of the feature maps after the second and third stages of MobileNetV3 are mixed between randomly paired images, with a mixing weight drawn from Beta(0.1, 0.1). It acts only during training and adds no parameters or inference cost. MixStyle is an existing technique; what is tested here is its combination with clean-teacher distillation.

What was not implemented. A feature-consistency loss between the two views, variety-adversarial training with a gradient-reversal head, and int8 quantisation were planned but not completed, so no result in this paper involves them.

V. EXPERIMENTAL SETUP

Models. Three ImageNet-pretrained networks from the timm library [24] serve as baselines: MobileNetV3-Large [4], MobileViT-S [6] and EfficientFormer-L1 [7]. The first is also the student in the distillation experiments.

Training. All runs use AdamW (learning rate 5e-4, weight decay 0.05), a one-cycle schedule with 10% warm-up, batch size 64, 224×224 inputs, mixed precision, and 12 epochs for PlantDoc and 8 for Paddy. The checkpoint with the best validation macro-F1 is kept and evaluated once on the test set. The same recipe is used for every model so that rows are comparable. Training ran on a single Colab T4 GPU. In the later runs (Tables IV and V and the MixStyle variety-shift runs) the teacher's outputs on the clean view were computed once and cached, which gives the same targets as running the teacher at every step.

Metrics. Accuracy and macro-F1 (the mean of per-class F1, so that rare classes count equally) are reported. Efficiency is measured as parameters, multiply-accumulate operations (GMACs, counted with PyTorch's FLOP counter at 224×224 and halved) and median CPU latency over 30 single-image forward passes on the Colab CPU. CPU latency varied by up to 10 ms between runs of the same network, so ranges are given. These are not on-device measurements.

Seeds. The PlantDoc ablation uses seeds 42, 43 and 44, and results are given as mean ± sample standard deviation. Baselines on Paddy and all variety-shift runs use seed 42 only. The PlantDoc teacher was trained once (seed 42) and reused.

Ablation rows. (a) baseline student; (b) plus field-style augmentation; (c) plus distillation with the baseline augmentation; (d) plus distillation and field-style augmentation.

Variety-shift protocol. Models are trained and validated on ADT45 images and tested on the 3,415 images of the nine other varieties (Section III).

VI. RESULTS AND ABLATION

Table I lists the three baselines, Table II gives the PlantDoc ablation, and Table III the in-distribution Paddy results. On PlantDoc, distillation lifts the MobileNetV3 student from 56.0 to 62.1 macro-F1 over three seeds (Table II). On the in-distribution Paddy split the models are within two points of each other, and the added components lower the score (Table III).

TABLE I
BASELINES (SEED 42): MOBILENETV3-LARGE [4], MOBILEVIT-S [6], EFFICIENTFORMER-L1 [7]; DATASETS PLANTDOC [1] AND PADDY DOCTOR [14]; ACCURACY / MACRO-F1 IN %

Model	Params (M)	GMACs	CPU (ms)	PlantDoc acc / F1	Paddy acc / F1
MobileNetV3-Large	4.2	0.215	20–29	56.8 / 55.4	93.9 / 93.4
MobileViT-S	4.9	1.416	82–91	72.9 / 71.1	93.6 / 93.2
EfficientFormer-L1	11.4	1.290	66–75	68.2 / 67.1	95.2 / 95.1

PlantDoc has 236 test images (margin of error about ±6 points); Paddy has 1,444 (about ±1.2 points).

As Table I shows, on PlantDoc both transformer models beat MobileNetV3 by 12–16 F1 points, a gap well above the test-set margin. On Paddy the three models are within two points. The transformers cost 6–7 times more operations and run three to four times slower on CPU.

TABLE II
PLANTDOC [1] ABLATION, MOBILENETV3 STUDENT [4]; DISTILLATION AFTER [8]

Configuration	Acc (%)	Macro-F1 (%)	F1 per seed (42 / 43 / 44)
(a) Baseline	56.9 ± 2.3	56.0 ± 1.7	55.4 / 57.9 / 54.6
(b) + field aug. (seed 42 only)	56.8	55.8	55.8
(c) + distillation	61.6 ± 0.5	61.1 ± 1.1	60.1 / 61.1 / 62.2
(d) + distillation + field aug.	**63.7 ± 0.9**	**62.1 ± 1.0**	**61.1 / 63.2 / 62.1**
Teacher MobileViT-S (seed 42)	72.9	71.1	71.1

Mean ± sample standard deviation over seeds 42, 43, 44.

In Table II, distillation beats the baseline on all three seeds, by 5 to 7 F1 points, and closes about 40% of the gap to the teacher ((62.1 − 56.0) / (71.1 − 56.0)). The student keeps MobileNetV3's cost of 0.215 GMACs against 1.416 for the teacher, which is 6.6 times fewer, and distillation adds nothing at inference time. Field augmentation alone changed nothing (55.8 against 55.4) and added only about one F1 point on top of distillation, within the seed spread, so the gain comes from distillation. The seed spread does not include the sampling error of a 236-image test set, so the result is consistent across seeds but not proven.

TABLE III
PADDY DOCTOR SUBSET [14], IN-DISTRIBUTION SPLIT (SEED 42)

Configuration	Acc (%)	Macro-F1 (%)
MobileNetV3 baseline	93.9	93.4
+ field augmentation	91.0	90.9
+ distillation + field aug.	90.1	89.8
Teacher EfficientFormer-L1	95.2	95.1

In Table III, field augmentation lowers macro-F1 by 2.5 points and distillation lowered it by a further 1.1, which is within the margin. Validation F1 of the augmented runs was still rising at epoch 8, and the teacher is only 1.7 F1 points better than the student, so there is little to distil. Whether longer training closes the gap was not tested, and these remain possible reasons rather than established ones.

Style mixing and tuning. Table IV adds MixStyle to the distilled student (baseline augmentation), and Table V sweeps the distillation weight α on one seed, choosing by validation F1 only.

TABLE IV
PLANTDOC: DISTILLATION WITH MIXSTYLE [23] (BASELINE AUGMENTATION)

Configuration	Acc (%)	Macro-F1 (%)	F1 per seed (42 / 43 / 44)
(c) distillation, α = 0.5	61.6 ± 0.5	61.1 ± 1.1	60.1 / 61.1 / 62.2
(e) + MixStyle, α = 0.5	63.4 ± 1.8	62.9 ± 1.8	61.2 / 62.8 / 64.8
(f) + MixStyle, α = 0.7	63.3 ± 1.5	62.1 ± 2.2	61.7 / 60.0 / 64.5

TABLE V
DISTILLATION WEIGHT A, PLANTDOC, SEED 42 (NO MIXSTYLE)

α	Val F1 (%)	Test acc (%)	Test F1 (%)
0.3	62.0	57.6	56.4
0.5	63.7	61.0	60.1
0.7	**66.7**	**64.4**	**62.8**
0.9	65.4	66.5	65.3

In Table IV, MixStyle with α = 0.5 beats plain distillation on all three seeds (by 1.1 to 2.6 F1 points), but with α = 0.7 the three seeds point in different directions (62.1 ± 2.2), and the differences among rows (c) to (f) are about one seed standard deviation. No gain from style mixing is therefore claimed. In the α sweep (Table V), validation F1 peaks at α = 0.7 while test F1 keeps rising to 65.3 at α = 0.9. Because the choice was to be made on validation, and because the sweep used one seed and a 236-image test set, α = 0.5 was kept for the main results, and heavier distillation checked across seeds is left as future work.

VII. ANALYSIS AND DISCUSSION

Models trained on one rice variety fail on the others (Table VI): accuracy falls from above 90% on ADT45 validation images to 34–42% on the nine unseen varieties.

TABLE VI
VARIETY SHIFT ON PADDY DOCTOR [14]: TRAIN AND VALIDATE ON ADT45, TEST ON 3,415 IMAGES OF THE OTHER VARIETIES (SEED 42)

Model	ADT45 val F1 (%)	Unseen-variety acc / F1 (%)
EfficientFormer-L1	**97.0**	**42.1 / 34.5**
MobileNetV3 baseline	93.0	37.7 / 31.5
+ field augmentation	89.0	34.1 / 28.9
+ field aug. + distillation (ADT45-only teacher)	88.6	35.1 / 29.3
+ MixStyle (baseline aug.)	93.0	35.6 / 30.1
+ MixStyle + distillation (ADT45-only teacher)	92.4	39.1 / 31.7

Chance is about 10% for ten classes.

In Table VI, the transformer holds up about 3 F1 points better than the CNN, a gap above the margin of error on 3,415 images (about ±1.7 points in accuracy). Field augmentation lowered the student's score by 2.6 F1 points, and distillation recovered only 0.3 of them. One possible reason, not tested here, is that the ADT45-only teacher itself scores only 34.5 F1 on the other varieties, so it cannot pass on robustness it does not have. The collapse also fits the data: disease and variety are correlated in the Paddy subset (Section III), so a model can partly learn the variety and the crop age instead of the lesion. Validation on ADT45 images cannot reveal this, because those images come from the same variety as the training data.

Style mixing did not help under variety shift. MixStyle alone scored 30.1 F1 against 31.5 for the baseline, and MixStyle with distillation scored 31.7 (accuracy 39.1 against 37.7). These differences are within the margin, and none of the student variants comes near the transformer's 34.5.

What the attention maps show. Fig. 1 shows Grad-CAM maps [25] of the baseline and distilled students on test images that the baseline misclassified and the distilled student classified correctly. Distillation does not make the maps more lesion-focused. On the apple rust leaf (row 1) the distilled student's highlight is broader and covers more of the upper leaf, although neither map sits on the individual spots. On the bell-pepper leaf spot image (row 5) the baseline highlights the brown lesion, while the distilled student moves its attention to the upper part of the leaf and a nearby pepper, yet still classifies the image correctly. In the three bell-pepper leaf images (rows 2 to 4) and the blueberry leaf (row 6) both students also respond to neighbouring leaves and background, and in rows 3, 4 and 6 the distilled student's maps spread further over them. Six examples are too few to say that distillation teaches the student where the disease is; what the maps do show is that cluttered field backgrounds draw attention away from the target leaf.

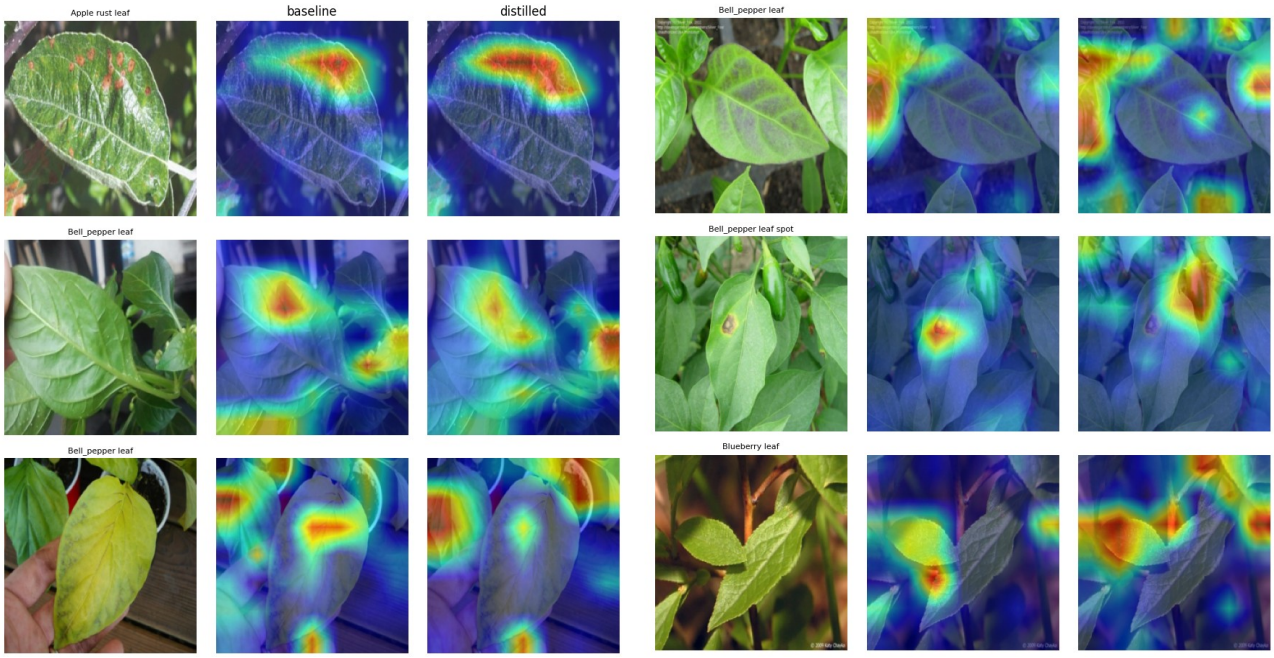


Fig. 1. Grad-CAM maps (last block of MobileNetV3-Large) for six PlantDoc test images that the baseline misclassified and the distilled student classified correctly. Each half shows, per row: input, baseline student, distilled student. Left half: rows 1–3; right half: rows 4–6.

Limitations.

- The PlantDoc test set has 236 images, and the Paddy comparisons and all variety-shift results use a single seed.
- The teacher was trained once. The temperature and epoch counts were not tuned, and α was swept on one seed only. Paddy runs were short (8 epochs).
- A 10,407-image subset of Paddy Doctor was used instead of the full 16,225-image release, and the images were downsized.
- CPU latency comes from a shared Colab CPU, not a phone, and no quantised or on-device measurement was made.
- The planned consistency loss, variety-adversarial training and int8 quantisation were not completed.
- PlantDoc and Paddy share no classes, so a direct cross-dataset test was not possible; the variety shift serves as a substitute.

Future work. Several directions follow from these results: a gradient-reversal head against the variety label, a feature-consistency loss between the clean and distorted views, int8 quantisation with on-device latency, larger distillation weights checked across seeds, variety-balanced training, and a per-class analysis of the variety-shift errors.

VIII. CONCLUSION

Distilling a MobileViT-S teacher into a MobileNetV3 student raised macro-F1 on PlantDoc field images from 56.0 ± 1.7 to 62.1 ± 1.0 over three seeds. That recovers about 40% of the teacher's advantage while keeping the student's 0.215 GMACs, 6.6 times fewer operations than the teacher. The gain comes from the distillation, while the field-style augmentation adds little. On the in-distribution Paddy split the three baselines are within two points of each other, and neither augmentation nor distillation helped. Under a change of rice variety every model collapses to 29–35 macro-F1, with the transformer about 3 points ahead of the CNN and no benefit from distillation or from MixStyle. Robustness to unseen varieties therefore remains open, and an in-distribution Paddy score should not be read as field readiness. Most results come from single seeds on small test sets, so they should be confirmed with more seeds, longer training and on-device measurements.

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